

Digital Logic Design

Midterm

2025/04/14

姓名: _____ 學號: _____ 系級: _____

- Convert the hexadecimal number DD42A to binary, and then convert it from binary to octal. (12%)
Notice: Both binary and octal are required.

$$\text{DD42A}_{16} = 1101_1101_0100_0010_1010 = (3352052)_8$$

- An n-bit signed binary number can be represented in the signed-2's complement, signed-1's complement, or signed magnitude representations. What are the ranges for each representation to represent an n-bit signed binary number?(18%)

signed-2's complement: $-(2^{(n-1)}) \sim (2^{(n-1)})-1$

signed-1's complement: $-((2^{(n-1)})-1) \sim (2^{(n-1)})-1$

signed magnitude: $-((2^{(n-1)})-1) \sim (2^{(n-1)})-1$

- Find the complement of the following expressions:(18%)

(a) $xyz + x'y'z'$

(b) $(x + y')(y + z')(z + x')$

(c) $w'x' + w(x + y + z)$

2.9 (a) $F' = (xyz + x'y'z')' = (xyz)'(x'y'z')' = (x' + y' + z')(x + y + z)$

(b) $F' = [(x + y')(y + z')(z + x')] = (x + y')' + (y + z')' + (z + x')' = x'y + y'z + z'x$

(c) $F' = [w'x' + w(x + y + z)]' = (w'x')'[w(x + y + z)]' = (w + x)[w' + (x + y + z)] = (w + x)(w' + x'y'z')$

- Given the Boolean functions F_1 and F_2 , show that
 - The Boolean function $E = F_1 + F_2$ contains the sum of the minterms of F_1 and F_2 .
 - The Boolean function $G = F_1 F_2$ contains only the minterms that are common to F_1 and F_2 .(12%)

2.10 (a) $F_1 + F_2 = \sum m_{1i} + \sum m_{2i} = \sum (m_{1i} + m_{2i})$

(b) $F_1 \cdot F_2 = \sum m_i \sum m_j$ where $m_i m_j = 0$ if $i \neq j$ and $m_i m_j = 1$ if $i = j$

5. Simplify the following Boolean function F, together with the don't-care conditions d, and then express the simplified function in sum-of-minterms form:(20%)

- (a) $F(x,y,z) = \Sigma(0,3,5)$
 $d(x,y,z) = \Sigma(1,6,7)$
- (b) $F(A,B,C,D) = \Sigma(1,2,4,6,12)$
 $d(A,B,C,D) = \Sigma(0,3,5,7,11,15)$
- (c) $F(A,B,C,D) = \Sigma(3,5,6,11)$
 $d(A,B,C,D) = \Sigma(4,7,9,12,15)$
- (d) $F(A,B,C,D) = \Sigma(4,5,8,9,14,15)$
 $d(A,B,C,D) = \Sigma(2,3,12,13)$

3.15 (a) $F(x,y,z) = \Sigma(0,3,5)$ $d(x,y,z) = \Sigma(1,6,7)$

		y			
		yz			
x	0	00	01	11	10
	0	m_0 1	m_1 x	m_3 1	m_2
x	1	m_4	m_5 1	m_7 x	m_6 x
	1				

$$F = x'y' + z$$

(b) $F(A,B,C,D) = \Sigma(1,2,4,6,12)$ $d(A,B,C,D) = \Sigma(0,3,5,7,11,15)$

		C			
		CD			
A	00	00	01	11	10
	00	m_0 x	m_1 1	m_3 x	m_2 1
	01	m_4 1	m_5 x	m_7 x	m_6 1
	11	m_{12} 1	m_{13}	m_{15} x	m_{14}
10	10	m_8	m_9	m_{11} x	m_{10}

$$F(A,B,C,D) = A' + BC'D'$$

(c) $F(A, B, C, D) = \Sigma(3, 5, 6, 11)$ $d(A, B, C, D) = \Sigma(4, 7, 9, 12, 15)$

$AB \backslash CD$		C			
		00	01	11	10
A	00	m_0	m_1	m_3 1	m_2
	01	m_4 x	m_5 1	m_7 x	m_6 1
	11	m_{12} x	m_{13}	m_{15} x	m_{14}
	10	m_8	m_9 x	m_{11} 1	m_{10}

D

$$F(A, B, C, D) = A'B + CD$$

(d) $F(A, B, C, D) = \Sigma(4, 5, 8, 9, 14, 15)$ $d(A, B, C, D) = \Sigma(2, 3, 12, 13)$

$AB \backslash CD$		C			
		00	01	11	10
A	00	m_0	m_1	m_3 x	m_2 x
	01	m_4 1	m_5 1	m_7	m_6
	11	m_{12} x	m_{13} x	m_{15} 1	m_{14} 1
	10	m_8 1	m_9 1	m_{11}	m_{10}

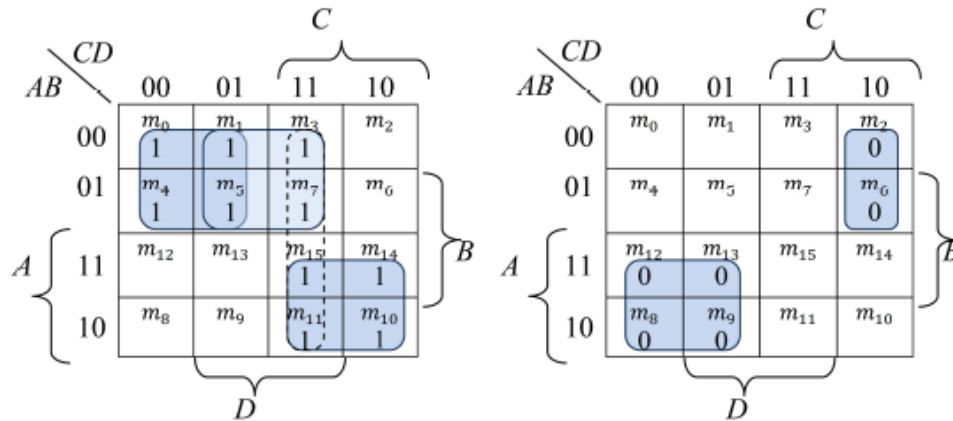
D

$$F(A, B, C, D) = AC' + BC' + AB$$

6. Give three possible ways to express the following Boolean function with a fewer number of literals:(10%)

$$A'B'C' + A'CD + A'C'D' + A'BD + ABC + AB'C$$

3.14



SOP form (using 1s): $F = A'C' + A'D + AC$

Alternative SOP form: $F = A'C' + AC + CD$

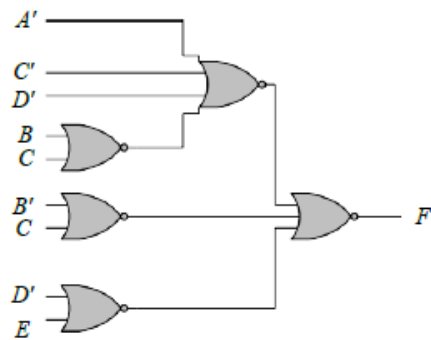
POS form (using 0s): $F' = AC' + A'CD'$

$$F = (A' + C)(A + C' + D)$$

7. Draw (a) the multiple-level NOR circuit for the following expression:
 $CD(B + C)A + (BC' + DE')$, and (b) repeat (a) for a NAND circuit.(10%)

3.20 Multi-level NOR:

$$\begin{aligned}
 F &= ACD(B + C) + (BC' + DE') \\
 F' &= [ACD(B + C) + (BC' + DE')]' \\
 F' &= [(A' + C' + D')(B + C) + (B' + C)' + (D' + E)]' \\
 F' &= [((A' + C' + D') + (B + C)')' + (B' + C)' + (D' + E)]' \\
 F' &= [(A' + C' + D' + (B + C)')' + (B' + C)' + (D' + E)]'
 \end{aligned}$$



Multi-level NAND:

$$\begin{aligned}
 F &= CD(B + C)A + (BC' + DE') \\
 F' &= [CD(B + C)A]' [BC' + DE']' \\
 F' &= [CD(B'C')'A]' [BC' + DE']' \\
 F' &= [CD(B'C')'A]' [(BC')' (DE')]'
 \end{aligned}$$

